COMPSCI 714 Hackathon-based project

Group Technical Report

Team	Team 14
Use Case	Immunization Guidelines Advisor Agent
Course	COMPSCI 714 AI Architecture and Design - S1 2026
Partners	University of Auckland & Microsoft

1. Project Overview

1.1 Executive Summary

The IMAC Immunization Advisor is a proof of concept AI tool built to help healthcare professionals find immunization information faster. In a clinical setting, staff often need to check vaccine schedules, eligibility rules, catch-up guidance, or contraindications, but the relevant information can be buried across long guideline documents.

Our project tries to solve this by creating a chat-based assistant that answers immunization questions using official IMAC source material. The system does not just rely on the language model by itself. Instead, it first searches the guideline documents, selects the most relevant sections, and then uses Azure OpenAI to generate a response based only on those sources.

The main goal of the proof of concept is to show that AI can support clinical workflows by saving time, improving consistency, and making it easier for users to verify where an answer came from. The tool is not intended to replace clinical judgement, but to act as a support system that gives source backed guidance.

1.2 Use Case Description

The selected use case is an immunisation guidance assistant for healthcare professionals. Healthcare staff often need quick answers about vaccine schedules, catch up plans, eligibility, side effects, contraindications, pregnancy, immunocompromised patients, and other special cases.
The challenge is that the correct answer is not always easy to find. Relevant information may be spread across different sections of a handbook or guideline document. This can slow down decision making, especially when the user is already managing a patient query or working under time pressure.
An AI based solution is suitable for this use case because the task is mainly about finding the right information, summarising it clearly, and showing where it came from. To make the system safer, our design uses retrieval augmented generation rather than a general chatbot approach. This means the model is grounded in the guideline documents and is instructed not to guess when there is not enough evidence.

1.3 Team Members / Roles

Name	Student ID	Primary Role	Email
ANIKETH KESHAVA RAO	148270981	Project Lead	akes297@AUCKLANDUNI.AC.NZ
ALEENA ANNA	776431272	Data & Problem Analyst	asur767@aucklanduni.ac.nz
KRISHNA PATIL	547429072	AI Governance Lead	kpat172@aucklanduni.ac.nz
NISHEETH BHOOSHAN GUPTA	140150096	Implementation Lead	ugnp116@aucklanduni.ac.nz
SAKSHI PATIL	514609253	Solution Architect	spt793@aucklanduni.ac.nz

Primary roles should include: Project Lead / PM, Data & Problem Analyst, Solution Architect, Implementation Lead, AI Governance Lead.

1.4 Use Case Owner / Stakeholders

Name(s)	Title / Department	Role in Project	Contact
Theo Brandt	Use Case Owner – UoA	Primary Client	T.brandt@auckland.ac.nz
Ming Cheuk	Microsoft	Technical Support	Ming.cheuk@auckland.ac.nz
Thomas Lacombe	Teaching Team – UoA	Academic Supervisor	Thomas.lacombe@auckland.ac.nz
Clinical Advisor	Clinical Advisor, IMAC, University of Auckland	End users, human-in-the-loop	N/A
Patient	N/A	Indirect Beneficiaries	N/A

2. Project Success Criteria

- *The system can answer common immunisation questions using the IMAC knowledge base.*
- *Each answer includes citations or source references so the user can check the original material.*
- *The system avoids making up answers when the source material is not strong enough.*
- *The retrieval process is fast enough for a practical chat-based workflow.*
- *Personal information in user queries is redacted before the query is processed.*
- *The system provides a confidence score so users can understand how reliable the answer is.*
- *The interface is simple enough for a non-technical user to try during a demo.*
- *The team can clearly explain the architecture, safety decisions, limitations, and possible future improvements.*

3. Assumptions, Constraints & Dependencies

3.1 Assumptions

- *The IMAC guideline content provided is suitable for answering the main types of immunization questions in this project.*
- *The expected users are healthcare professionals or people working in a supervised healthcare context.*
- *Most questions in the project scope can be answered from the available documents.*
- *A two-stage retrieval approach will provide better results than basic keyword search alone.*
- *Some clinical details, such as age, should be kept because they are important for immunization advice.*
- *Direct personal identifiers should be removed before the query is sent through the AI pipeline.*

3.2 Constraints

- The project was completed within a hackathon timeframe, so the focus was on a working proof of concept rather than a production-ready system.
- The system has not gone-through formal clinical validation.
- The knowledge base is static, so the system can only answer based on the documents currently loaded.
- The current vector store is lightweight and suitable for a prototype, but not necessarily suitable for large-scale production use.

- The system depends on access to Azure OpenAI credentials and the correct local setup.

3.3 External Dependencies

- A chat interface where users can ask immunization-related questions.
- A Fast-API backend to manage requests and run the AI pipeline.
- Document ingestion and chunking of IMAC guideline content.
- Bi-encoder retrieval to quickly find candidate source chunks.
- Cross-encoder re-ranking to improve the relevance of retrieved chunks.
- Azure Open-AI based answer generation using only retrieved evidence.
- Source citations included with responses.
- Confidence scoring for the final answer.
- PII redaction before query processing.
- Query classification by vaccine type, clinical scenario, and caller type.
- Chat session storage, loading, and deletion.

4. Scope of Work – Technical Project Plan

4.1 In Scope

- *A chat interface where users can ask immunisation-related questions.*
- *A FastAPI backend to manage requests and run the AI pipeline.*
- *Document ingestion and chunking of IMAC guideline content.*
- *Bi-encoder retrieval to quickly find candidate source chunks.*
- *Cross-encoder reranking to improve the relevance of retrieved chunks.*
- *Azure OpenAI based answer generation using only retrieved evidence.*
- *Source citations included with responses.*
- *Confidence scoring for the final answer.*
- *PII redaction before query processing.*
- *Query classification by vaccine type, clinical scenario, and caller type.*
- *Chat session storage, loading, and deletion.*

4.2 Out of Scope

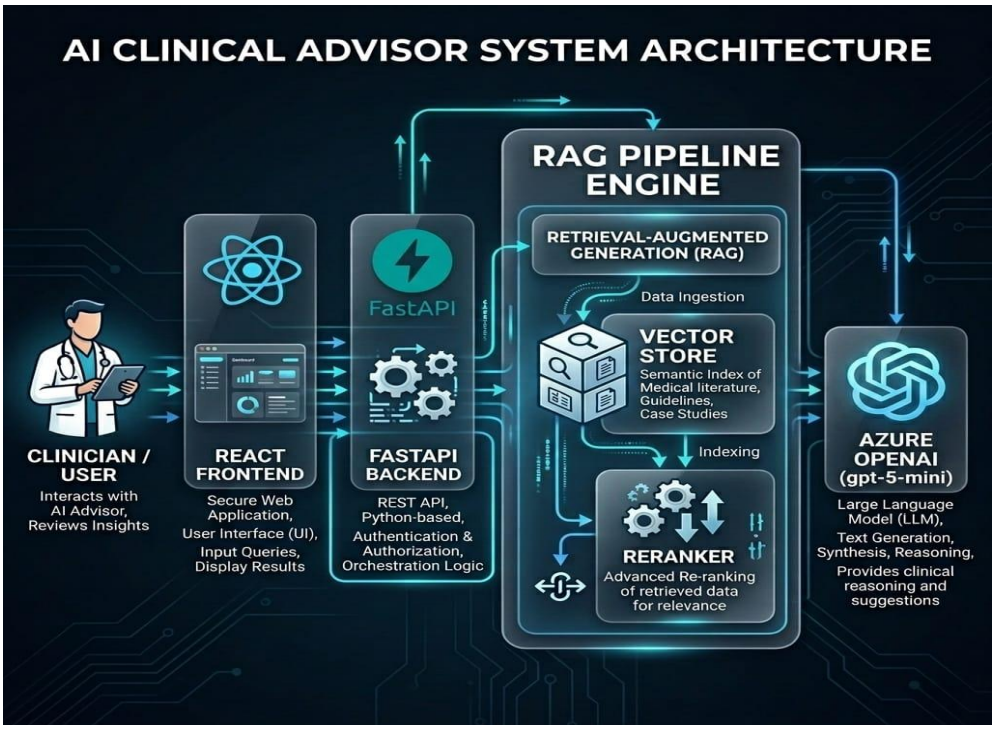
- Production deployment in a real healthcare environment.
- Formal medical device certification.
- Integration with patient records, hospital systems, or live NHI services.
- Automatic updating from external guideline websites.
- Full clinical validation by domain experts.
- Advanced analytics dashboards.
- Mobile app or voice interface.
- Complete security hardening and monitoring

5. Solution Architecture

5.1 High-Level Architecture Description

- *The system is made up of a React frontend, a FastAPI backend, and a retrieval-based AI pipeline.*
- *The user interacts with the system through a chat interface. When a question is submitted, the frontend sends the query to the backend. The first step in the backend is PII redaction, where personal information such as NHI numbers, phone numbers, emails, dates of birth, names, and addresses is masked.*
- *After this, the query is classified to identify details such as vaccine type, clinical scenario, and caller type. This classification makes the system more useful for reporting, triage, and understanding the types of questions being asked.*
- *The main AI process uses a two stage retrieval pipeline. First, a Bi-encoder model searches the knowledge base and returns the top candidate chunks. Then, a Cross-encoder reranks those chunks to select the most relevant evidence. The best chunks are passed to Azure OpenAI, which generates the final answer.*
- *The response shown to the user includes the answer, citations, a confidence score, classification information, and PII redaction metadata. Chat sessions are stored in PostgreSQL so users can load previous conversations or delete them.*
- *The architecture was designed this way to keep the system grounded, transparent, and safer than a general purpose chatbot.*

5.2 Architecture Diagram



5.3 Azure AI Foundry Services Used

Azure Service / Tool	Purpose in Solution	Justification
Azure OpenAi Service	Generates the final clinical response using the retrieved IMAC guideline evidence through the gpt-5-mini model.	Chosen because it provides enterprise-grade large language model capabilities with strong reliability, scalability, and integration support for healthcare-oriented AI solutions. The model is used in a grounded RAG setup to minimise hallucinations and ensure answers are based only on retrieved source material.

6. Milestones & Deliverables

Week	Activity / Milestone	Deliverable	Due Date	Owner / Notes
Week 1	Project planning and requirement analysis	Initial project scope, architecture discussion, and team role allocation	Week 1	Project Lead – Team coordination and planning
Week 1	IMAC guideline data collection and preprocessing	Parsed and structured guideline document	Week 1	Data & Problem Analyst
Week 2	Backend API setup using FastAPI	Functional FastAPI backend with query endpoints	Week 2	Implementation Lead
Week 2	Frontend React interface development	Chat-based clinical UI prototype	Week 2	Solution Architect
Week 3	Bi-Encoder retrieval implementation	Top-k candidate retrieval pipeline using all-MiniLM-L6-v2	Week 3	AI/Implementation Team
Week 3	Cross-Encoder reranking integration	Precision reranking pipeline using ms-marco-MiniLM-L-6-v2	Week 3	Implementation Lead
Week 4	Azure OpenAI integration	Grounded answer generation using gpt-5-mini	Week 4	Solution Architect
Week 4	Testing, debugging, and optimisation	Stable proof-of-concept system	Week 4	Entire Team

7. Risk Management

Risk	Likelihood	Impact	Mitigation Strategy
Retrieval returns irrelevant chunks	Low	High	Implement two-stage retrieval with cross-encoder reranking to improve precision and relevance.
Poor system performance or latency	Low	Medium	Use lightweight NumPy vector storage and optimise retrieval stages for fast response times.
Static knowledge base becomes outdated	Medium	Medium	Clearly state that responses are limited to currently loaded IMAC guideline documents.
Lack of formal clinical validation	High	High	Present the system strictly as a proof-of-concept advisory tool rather than a production clinical system.

8. Work Plan & Responsibilities

Task	Owner	Contributor(s)	Target Date	Actual Date
Project planning and coordination	ANIKETH KESHAVA RAO	Entire Team	Week 1	Week 1
Requirement analysis and use-case definition	ALEENA ANNA	ANIKETH KESHAVA RAO	Week 1	Week 1
System architecture design	SAKSHI PATIL	NISHEETH BHOOSHAN GUPTA	Week 1	Week 1
FastAPI backend implementation	NISHEETH BHOOSHAN GUPTA	ANIKETH KESHAVA RAO	Week 2	Week 2
React frontend development	SAKSHI PATIL	ALEENA ANNA	Week 2	Week 2
PII redaction engine	KRISHNA PATIL	SAKSHI PATIL	Week 3	Week 3
Query classification module	ALEENA ANNA	KRISHNA PATIL	Week 3	Week 3
Confidence scoring and citations	ANIKETH KESHAVA RAO	NISHEETH BHOOSHAN GUPTA	Week 3	Week 3
Testing and debugging	Entire Team	Entire Team	Week 4	Week 4
Documentation and final presentation	Entire Team	Entire Team	Week 4	Week 4

9. Use of Generative AI

GenAI Tool Used	Purpose / Task	Team Member(s)	Notes
e.g., Claude / ChatGPT	Draft project plan sections	All	Reviewed and edited by team
e.g., GitHub Copilot	Brainstorming documentation structure and wording improvements	ALEENA ANNA	Content manually verified before inclusion
Claude Code	Backend and frontend coding assistance	NISHEETH BHOOSHAN GUPTA, ANIKETH KESHAVA RAO	Used for code suggestions and productivity support

10. AI Governance

The IMAC Immunisation Advisor was developed using a safety-first and governance-focused approach appropriate for a healthcare support system. The project follows Retrieval-Augmented Generation (RAG) principles to ensure that responses are grounded in official IMAC immunisation guideline documents rather than relying solely on the language model's internal knowledge.

To protect user privacy, all incoming queries pass through a PII redaction layer before processing. Sensitive information such as NHI numbers, phone numbers, names, email addresses, dates of birth, and street addresses are automatically masked. However, clinically relevant details such as age or gender are preserved because they are important for immunisation reasoning.

The system improves transparency by providing:

- Source citations for every response
- Confidence scores indicating retrieval reliability
- Query classification metadata
- Explicit refusal responses when evidence is insufficient

The project acknowledges several limitations:

- The system has not undergone formal clinical validation
- Responses are limited to the currently loaded IMAC documents
- Retrieval quality depends on document chunking and reranking accuracy
- The system is intended as an advisory support tool and not a replacement for clinical judgement

Potential risks such as hallucinations, misinformation, or retrieval failure are mitigated using:

- Two-stage retrieval architecture
- Cross-encoder reranking
- Strict prompting instructions
- No-speculation response policy

The team also considered ethical concerns around AI use in healthcare, particularly the importance of explainability, user trust, and avoiding overreliance on automated systems. Users are encouraged to verify answers directly against the cited guideline material before making clinical decisions.

11. Resources & References

- Azure OpenAI Service Documentation
- Azure AI Foundry Documentation
- FastAPI Documentation
- React Documentation
- PostgreSQL Documentation
- IMAC Immunisation Handbook and supporting guideline documents
- Course material from COMPSCI 714 AI Architecture and Design
- Internal project architecture diagrams and workflow documentation

Appendix A: Document Version History

Version	Date	Changes Made	Contributor(s)
v0.1	Week 1	Initial project template and scope definition	Entire Team
v0.2	Week 2	Added architecture design and technical components	ANIKETH KESHAVA RAO, NISHEETH BHOOSHAN GUPTA
v0.3	Week 3	Added RAG pipeline, retrieval design, and Azure integration details	ANIKETH KESHAVA RAO
v0.4	Week 4	Added governance, PII redaction, and risk management sections	ALEENA ANNA